Home(y): On-Demand Service Dispatch System

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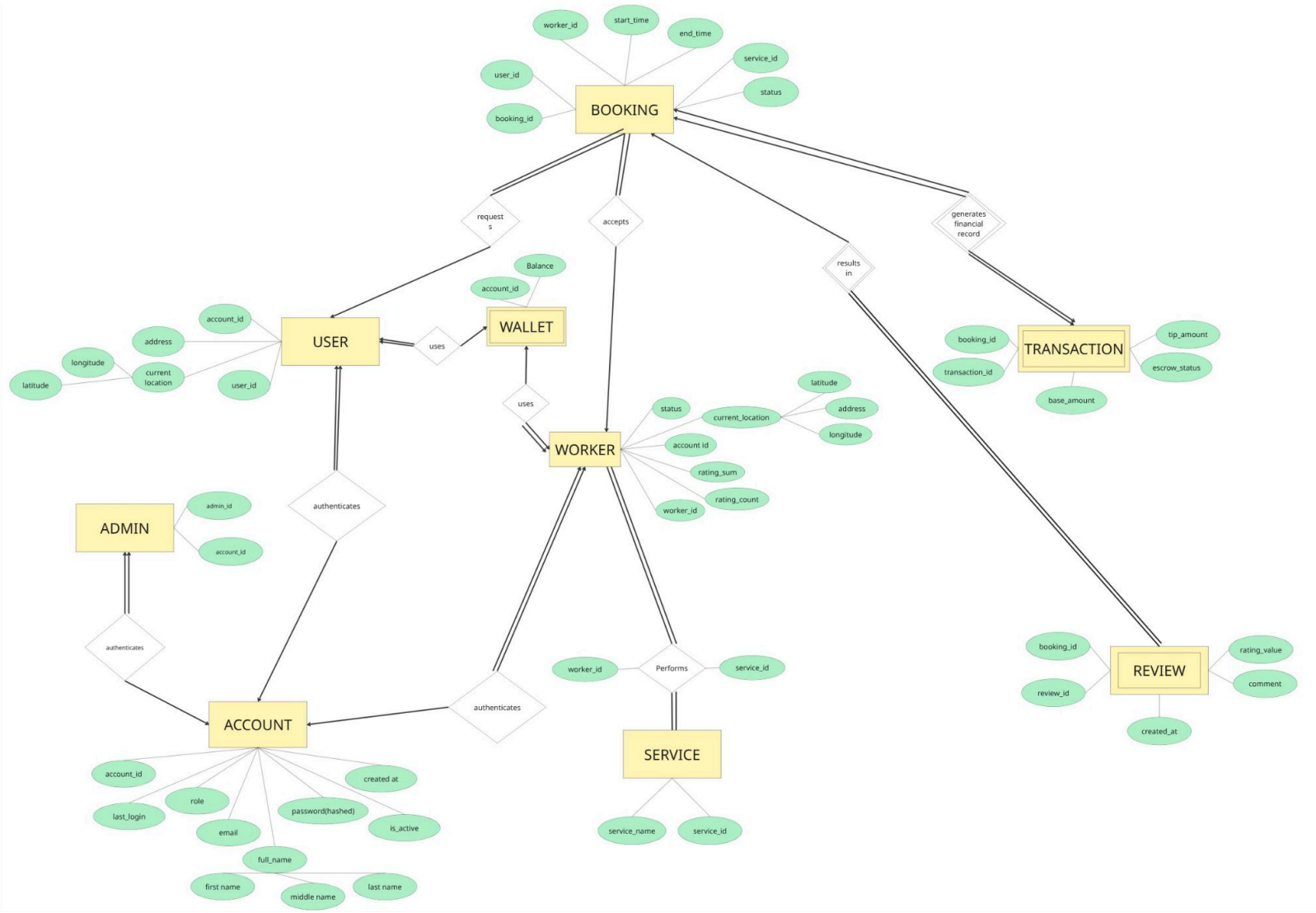
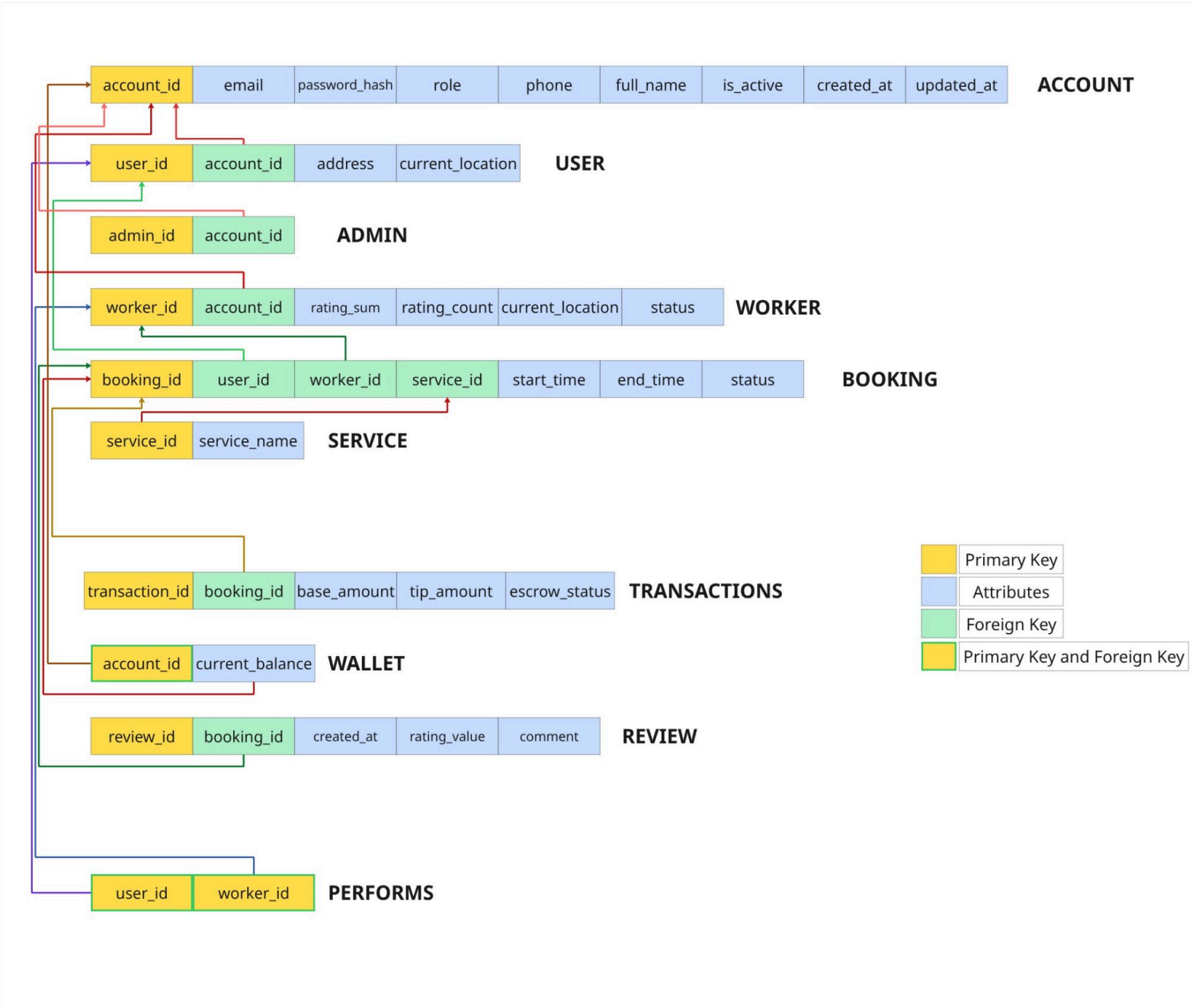
Entity-Relationship Diagram



Miro Link: [LINK]

Relational-Model



Miro Link: [LINK]

List of entity sets and their attributes

- **BOOKING**
  - Attributes: user_id, booking_id, worker_id, start_time, end_time, service_id, status
  - **Primary Key**: booking_id
  - **Foreign Keys:** user_id, worker_id, service_id
  - **Entity Type:** Strong

- **USER**
  - Attributes:user_id, account_id, current_location(longitude,latitude), address
  -Primary Key: user_id
  -Foreign Key: account_id
  -Entity Type: Strong

- **ADMIN**
  - Attributes:admin_id, account_id
  -Primary Key: admin_id
  -Foreign Key: account_id
  -Entity Type: Strong

- **WORKER**
  - Attributes: worker_id, account_id, rating_sum, rating_count, current_location, status
  -Primary Key:worker_id
  -Foreign Key:account_id
  Entity Type: Strong

- **BOOKING**
  - Attributes:booking_id, user_id, worker_id, service_id, start_time, end_time, status
  -Primary Key:booking_id
  -Foreign Key:account_id
  -Entity Type: Strong

- **SERVICE**
  - Attributes:service_id, service_name
  -Primary Key:service_id
  -Foreign Key: N.A.
  -Entity Type: Strong

- **TRANSACTION**
  - Attributes: transaction_id, booking_id, base_amount, tip_amount, escrow_status
  -Primary Key:transaction_id
  -Foreign Key:,booking_id

-Entity Type: Weak

Reason for being a weak entity: The entity Transaction is related to another entity known as Booking and is labelled weak in nature as in the absence of a booking, the transaction cannot exist on its own and is hence dependent on booking.

- **WALLET**
  -Attributes: user_id,current_balance
  -Primary Key: account_id
  -Foreign Key:account_id
  -Entity Type: Weak

  Reason for being a weak entity: The entity Wallet is related to entities worker and user and cannot exist in their absence as if there are no users and workers then there will be no instances of any wallets in the wallets entity sets

- **REVIEW**
  - Attributes: review_id,booking_id,created_at,rating_value,comment
  -Primary Key: account_id
  -Foreign Key: booking_id
  -Entity Type: Weak

  Reason for being a weak entity: The entity Review is related to the entity booking and cannot exist in its absence as without any instance of a booking, there cannot be any corresponding instance of a review.

- **PERFORMS**
  - Attributes: user_id,worker_id
  - Primary Key:user_id+worker_id
  - Foreign Key: user_id, worker_id
  - Entity Type: Weak

  Reason for being a weak entity: The entity is a combination of attributes user_id and worker_id which are attributes of worker and users and in their absence it cannot exist on its own hence making it dependent and weak.

Relationships and Cardinalities

The following relationships define how entities interact, following the roles and constraints of the system.

<u>One-to-One Relationships (1:1)</u>

- **Authenticates**: An ACCOUNT authenticates exactly one USER, WORKER, or ADMIN.
- **Results** In: A BOOKING results in exactly one REVIEW.
- **Uses**: A USER possesses exactly one WALLET.

<u>One-to-Many Relationships (1:N)</u>

- **Requests**: A USER can request multiple BOOKINGS, but a single Booking belongs to only one User.
- **Generates**: A BOOKING generates exactly one TRANSACTION.
- **Accepts**: A WORKER can accept multiple BOOKINGS over time, but a single Booking is assigned to only one Worker.

<u>Many-to-Many Relationships (M:N)</u>

- **Performs**: A WORKER can perform multiple SERVICES, and a specific SERVICE type can be performed by multiple Workers. This is resolved via the PERFORMS table.

<u>Ternary Relationships</u>

User – Worker – Service These three entities are associated together using the **Booking** relation. A booking represents an event where a specific User hires a specific Worker to perform a specific Service.

This ternary relationship can be conceptually broken down into the following binary interactions, though they are tightly coupled in the BOOKING entity:

- **User – Worker:** A user hires a worker.
- **Worker – Service:** A worker performs a service type.
- **User – Service:** A user requests a specific type of service.

CONTRIBUTIONS:

With discussions and meets we together made up the ER Model and Relational Model for this deadline.